SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED
CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:
I Kiruthika K (1924260011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
Iunderstand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

Kiruthika K (1924260011)

Analytical ProblemSolving

1. Solve the Water Jug problem: youaregiven2jugs, a 4-gallon one and 3- gallon one. Neither has any measuringmakeron it. There is a pump that can be used to fill the jugs with water.Howcanyouget exactly 2 gallons of water into 4-gallon jug? Explicit assumptions:Ajugcan be filled from the pump, water can be poured out of a jugontotheground, water can be poured from one jug to another and that therearenoothermeasuring devices available.

Problem: We have two jugs: a 4-gallon jug and a 3-gallon jug. We need to measure exactly 2 gallons using these jugs.

Solution: Step-by-Step

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour water into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water into the 4-gallon jug until it is full	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining water into the 4-gallon jug	2	0

Result: Exactly 2 gallons of water are measured in the 4-gallon jug.

Assumptions

1. Both jugs have fixed capacities.
2. Water is available in unlimited quantities.
3. Water can be poured between jugs or emptied onto the ground.
4. Nomeasuring devices are available.

Conclusion

Thewater jug problem can be solved using a **state-space search approach**,where each state represents the amount of water in the two jugs.

2.Findout howMarsRover, analyse and explore the samples on Mars by answering thefollowing questions.

i. What are thepercept's for this agent?
ii. Characterizethe operating environment.
iii. What are theactions the agent can take?
iv. How can oneevaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent? Why?

AMars Rover is an autonomous robot designed to explore the surface of Mars, collect samples, and send information to Earth.

i. What are the percepts for this agent?

Percepts are the information received by an agent through its sensors.
The Mars Rover receives the following percepts:

- Images of the Martian surface.
- Temperature and atmospheric conditions.
- Distance to rocks and obstacles.
- Soil and rock composition.
- Rover position and movement.
- Battery level and system status.

ii. Characterize the operating environment.

The Mars Rover operates in the following environment:

- **Partially observable:** It cannot observe the entire Martian surface at once.
- **Single-agent:** It performs its exploration independently.
- **Stochastic:** Some outcomes, such as wheel movement on loose soil, are uncertain.
- **Sequential:** Each action affects future actions.
- **Dynamic:** Environmental conditions may change.
- **Continuous:** Position, distance, temperature, and movement can vary continuously.

iii. What are the actions the agent can take?

The Mars Rover can perform these actions:

1. Move forward and backward.
2. Turn left and right.
3. Capture images.
4. Collect soil and rock samples.
5. Analyze samples using scientific instruments.
6. Avoid obstacles.
7. Communicate data to Earth.
8. Recharge or conserve battery power when possible.

iv. How can one evaluate the performance of the agent?

The performance of the Mars Rover can be evaluated using:

- Number of samples collected successfully.
- Accuracy of scientific data.
- Distance explored.
- Successful obstacle avoidance.
- Efficient battery usage.
- Amount of useful data transmitted to Earth.
- Safety and reliability of operations.

v.Whattypeofagentarchitectureismostsuitable?Why?

Answer: Model-Based Utility Agent
Amodel-basedutilityagentissuitablebecause the Mars Rover operates in a partially observable anduncertainenvironment.
Reasons:

1. It maintainsaninternalmodelofthe environment.
2. It usessensordatatoupdateitsunderstanding.
3. It comparespossibleactionsbased on their benefits.
4. It selectsactionsthatimprovescientific results while considering safety, time, and energy.

Conclusion: Amodel-basedutilityagent helps the Mars Rover make intelligent decisions andexploreMarsefficiently.

3. Explorethesearchspace using search strategies and formulate the problem componentsforthe8-Queens problem using the following information: Place 8 queensonachessboard such that none of the queen's attack any of the others.Aconfigurationof 8 queens on the board as shown in the figure below, but thisdoesnotrepresent a solution as the queen in the first column is on the samediagonalasthequeen in the last column.

Problem:

Placeeight queens on an 8 × 8 chessboard so that no two queens attack each other.

Rules

- No two queens can be in the same row.
- No two queens can be in the same column.
- No two queens can be on the same diagonal.

Search Strategy

The **Backtracking Search Algorithm** is suitable for solving the Eight Queens Problem.

Algorithm

1. Startwith an empty chessboard.
2. Place aqueen inthefirstrow.
3. Check whether theposition is safe.
4. If safe, movetothenextrow.
5. Ifnosafepositionexists,returntothe previous row and move that queen.
6. Continueuntilalleightqueensareplaced safely.

Pseudocode

```
N = 8
FUNCTION Solve(row):
    IFrow== N:
       PRINT board
       RETURN TRUE

    FORcol= 0TO N-1:
       IF position(row,col) is safe:
          Place queen at(row,col)

          IF Solve(row+ 1):
             RETURNTRUE

          Removequeenfrom(row,col)

    RETURNFALSE
```

BEGIN:
Create an empty 8 × 8 board
Solve(0)
END
OneValid Solution

The following table gives the column position of the queen in each row.

Row	Column
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

The correspondingcolumn sequence is:
[1,5,8,6,3, 7,2,4]
Conclusion
Backtracking systematically explores possible queen positions and removes choices that lead to conflicts. This reduces unnecessary searching and finds a valid solution.

4. Acustomerwants totravelfromthe onelocationtoanother location using OLA Cab bookingmobileapplication.The customercanselect the any of the cab type as mini,micro,sedan,shared, prime andsoonbased on his comfort to reach the destination.Identify what type of problem-solving agent can be used and write thepseudocodefor the above problem.

Problem: A customer wants to travel from one location to another using OLA Cab. The customer can choose a mini, micro, sedan, or prime cab based on comfort and price.
Type of Problem-Solving Agent

A
Goal-BasedAgentwith Cost Optimization is suitable.
The agent identifies the customer's destination and selects a cab that satisfies the customer's requirements while minimizing the cost or travel time.
Steps Involved

1. **Initial state:** Customer's current location.
2. **Goal state:** Customer reaches the destination.
3. **Actions:** Select cab, book cab, and travel to the destination.
4. **State space:** Available locations, routes, and cab options.
5. **Path cost:** Cab fare, travel time, and other relevant costs.
6. **Goal test:** Check whether the customer has reached the destination.
7.

Pseudocode

BEGIN

INPUT source
INPUT destination
INPUT preferred_cab
INPUT budget

Get available cabs

```
Getestimated fares and travel times

FOReach cab:
   IFcab matches customer preference
     AND fare <= budget:
        Add cab to suitable_cabs

IFsuitable_cabs is empty:
   PRINT "No suitable cab available"

ELSE:
   Select the best cab based on fare and travel time
   Display cab details and estimated fare

   IFcustomer confirms booking:
      Book the selected cab
      Navigate to destination
      PRINT "Booking successful"

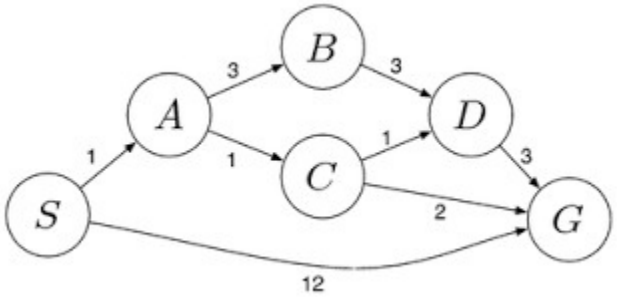
END
```

Conclusion

A goal-based agent with cost optimization helps customers choose a suitable cab based on their preferences, budget, and estimated travel time.

5.Alogisticscompanyoperatesadeliverynetwork where packages must be transportedbetweenwarehousesatminimum cost. Each route between warehouseshasanassociatedtravelcost(fuel + time). The company wants to determinetheleast-cost pathfromthestarting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The deliverynetwork is representedasaweighted graph:



Problem: A logistics company needs to transport packages from the starting warehouse **S** to the destination warehouse **G** at minimum cost.

Uniform Cost Search expands the path with the lowest cumulative cost first.

Given WeightedGraph

The edges and costs shown in the figure are:

Route	Cost
S → A	1
S → G	12
A → B	3 1
A → C	3 1
B → D	
C → D	

C → G	2
D → G	3

Step-by-Step UCSSolution

Start atS withcost 0.

Step 1: Available paths:

- S → A =1
- S → G =12

ChooseS → **A**,becauseitscostis1.

Step 2: ExpandA.

New paths:

- S → A →B=1+3=4 S →
- A →C=1+1=2 S → G
- =12

ChooseS → **A** → **C**,withcost2.

Step 3: ExpandC.

New paths:

- S → A →C→D=1+1+1=3 S → A
- →C→G=1+1+2=4

ChooseS → **A** → **C** → **D**,withcost3.

Step 4: ExpandD.

New path:

- S → A →C→D→G=1+1+1+3=6

The lowest-costremainingpathtothegoal is now S → A → C → G, with cost 4.

Step 5: SelectGwithcost4.Sincethedestination is reached with the lowest cumulative cost, thesearchterminates.

Final Answer

OptimalPath:

S → A → C→G

TotalCost:

Total Cost =1+1+2

Minimum Cost = 4

Conclusion

Uniform Cost Search finds the least-cost route by expanding paths in increasing order of cumulative cost. The optimal route from warehouse S to warehouse G is **S→A→C→G**, with a total cost of **4 units**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem